A Review of the Filtering Techniques used in EEG Signal Processing

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Abstract-In this paper, a general overview of the different kinds of filters, their applications in real world and the various pitfalls of filtering have been briefly discussed with a special focus on the use of various filtering techniques and their shortcomings in Electroenceplalogram(EEG) signal processing.

Keywords—Electroencephalogram(EEG), Signal-to-noise ratio(SNR), Gaussian filter, Butterworth filter, Chebyshev filter, Elliptic filter, Bessel filter, Impulse Response(FIR) filter, Infinite Impulse Response(IIR) filter, Information-centric social networks(IC-SN), Lifting Filter using Second Generation Wavelets (LFSGW)

I. INTRODUCTION

A filter is an AC circuit that is used in signal processing which passes or even amplifies particular frequencies and attenuates others. Thus, a filter can extricate frequencies of interest from signals containing undesirable or irrelevant frequencies called noise. The biggest challenge faced in signal processing is that the measured signals are often affected by some noisy artifacts, say, for example, noise that is produced by the environment, noise produced by instruments or noise produced by some sources which are not relevant to the experiment. However, if the useful signals and interferences occupy different regions of the spectrum, it is often possible to improve the signal-to-noise ratio or SNR by means of filtering the data.

II. NEED FOR FILTERS

Filters attenuate undesirable frequencies called noise from the applied signal, amplify desirable ones or perform both, say, for instance, a current component (DC) or slow fluctuation might be eliminated by using high-pass filters while power line components are diminished by using notch filters set at 50 or 60 Hz, and undesired high frequency elements might be eliminated by a technique called "smoothing" with the help of low pass filters. Filtering makes use of the difference in between the spectrum occupied by noise and that of the target signal to enhance SNR. It attenuates those signals in the part of the spectrum predominated by noise, more than those in the part containing the target signals.

III. AVAILABLE FILTERING TECHNIQUES

A. Gaussian filter: Gaussian filters have a Gaussian function (or a close approximation to it since the impulse response of an ideal Gaussian function is infinite) as its impulse response. A Gaussian filter does not have any overshoot if a step function is used as input. Furthermore, it has minimal rise and fall time. This is because it has minimal group delay.[1] Such a filter uses convolution with a Gaussian function to modify the input signal. A Gaussian filter of one dimension has the following impulse and frequency responses respectively:

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}\ ;\ \hat{g}(f) = e^{-\frac{f^2}{2\sigma_f^2}} \quad \text{where } \sigma\sigma_f = \frac{1}{2\pi} \tag{1}$$

where σ and σ_f denote standard deviation and the same for frequency domain respectively.

B. Butterworth filter **:** A Butterworth filter is a filter with a frequency response which has maximum flatness in the passband (devoid of ripples) and gradually reduces to nil in the stopband.[2] The gain for a Butterworth low pass filter expressed using transfer function H(jw) is:

$$G^2(w) = |H(jw)|^2 = \frac{G_0^2}{1+\left(\frac{jw}{jw_c}\right)^{2n}} \tag{2}$$

where n is the order of the filter, G(w) is the gain, w_c denotes the cutoff frequency which is approximately -3 dB while G_0 represents the DC gain or the gain at frequency zero.

C. Chebyshev (Type 1)Filter : Chebyshev Type 1 filter is one which shows a much more rapid roll-off as opposed to a Butterworth filter, along with the occurrence of ripple in the passband.[3] The gain (or amplitude) response $G_n(w)$ for the Chebyshev type 1 low pass filter of order n is given as:

$$G_n(w) = |H_n(jw)| = 1/\sqrt[2]{1 + \epsilon^2 T_n^2(\frac{w}{w_0})} \tag{3}$$

where w denotes the angular frequency, $\boldsymbol{\epsilon}$ denotes the ripple factor, w_0 indicates the cutoff frequency while T_n represents the nth order Chebyshev Polynomial ($T_n(\cos\Theta) = \cos(n\Theta)$).

D.Chebyshev (Type 2) Filter **:** Commonly called an inverse Chebyshev filter, the Chebyshev filter of type 2 is not usually used for it rolls off much slowly than a Chebyshev Type 1 filter; further, it requires much more components. It does not have passband ripple, but has equiripple in its stopband. [4] The gain is :

$$\frac{1}{\left(1+\frac{1}{\epsilon^2 T_n^2\left(\frac{w_0}{w}\right)}\right)^{\left(\frac{1}{2}\right)}} = \left(\frac{\epsilon^2 T_n^2\left(\frac{w_0}{w}\right)}{1+\epsilon^2 T_n^2\left(\frac{w_0}{w}\right)}\right)^{\frac{1}{2}} \tag{4}$$

where $\boldsymbol{\epsilon}$ denotes the ripple factor, w_0 indicates the cutoff frequency while T_n represents the nth order Chebyshev Polynomial ($T_n(\cos \Theta) = \cos(n\Theta)$).

E. Elliptic Filter **:** For an elliptic filter, there exists equalized ripple in both the passband as well as the stopband. The number of ripples in the bands can be adjusted independent of each other, and there does not exist any other filter of a given order for which the transition in gain from passband to stopband is quicker, for a particular value of the ripple factor, irrespective of whether the filter has equalized ripple or not. The gain for an Elliptic filter (low pass filter) is

$$G_n(w) = 1/\sqrt[2]{1+\epsilon^2 R_n^2(\xi, \frac{w}{w_0})} \tag{5}$$

Here, w is the angular frequency, R_n denotes the elliptic rational function of order n, also called the Chebyshev rational function, w_0 represents the cutoff frequency, ϵ the ripple factor while ξ the selectivity factor.[5]

F. Bessel Filter: A Bessel filter has a group delay with maximal flatness and retains the shape of signals post filtering in the passband. The Bessel filter somewhat resembles a Gaussian filter. Unlike a Gaussian filter whose time-domain step response has no overshoot, a Bessel filter displays a miniscule degree of overshoot which is still considerably insignificant in comparison to other filters like the Butterworth filter. The impulse response for a Bessel filter tends towards that of a Gaussian filter as the filter order increases.[6]

IV. Applications Of Filtering

A. Stock Market: In stock market, a filter is a technique for eliminating investment candidates. Filters enable investors to choose investments from a list of potential candidates, thereby saving time. Another major application in the field of finance is revealing trends from time series data by implementing the concept of moving averages which helps us deduce information and make forecasts about market trends. For example, the financial data obtained from Twitter and Yahoo can be used to forecast stock market values. Kalman filter can be implemented to reduce errors in the data by smoothening the noise manifesting as sudden peaks.[7] Daily stock trends can be forecasted using multi-filter feature selection.[8]

B. Social Media: Social media sites use filtering to rate and personalize content. The social networks created by social media users clearly exhibit their preferences and interests. These are used by the social media sites to filter the huge amount of content continuously generated to find ones that will interest the targeted viewer group. Social media sites thus facilitate novel ways of interacting with information-a phenomena known as social browsing.[9]

C. Image Processing: Filtering of images is a tool used for modifying or improving the quality of an image. For instance, images are filtered to accentuate certain features or do away with others. The chief image processing steps that implement filtering techniques are smoothing, sharpening and enhancement of edges. Application of filters increases the contrast by changing the colors of the pixels. The principal purpose of using filters is reducing the noise around characters thereby producing a high success rate of OCR (optical character recognition). [10]

D. Brain Science: Digital filtering is an important preprocessing operation to be performed before brain EEG data is analyzed. The usual practice is applying a high-pass filter for removing frequencies lower than 0.1 Hz or sometimes even 1 Hz and again applying a low-pass filter for attenuating frequencies greater than 40 or 50 Hz. Kernel-based filtering has been recently developed. A bilateral filter generates the higher frequency constituents of the original image. Then the original image's intensity component is approximately measured. Lastly, side window filtering technique is applied so as to minimize possible differences in intensity between the original image and the one obtained after being filtered by a low-pass filter. [11]

E. Acturial Science: In Acturial science, many popular credibility models can be formulated using Kalman filters to produce recursive premium forecasts like recursive prediction errors.[12]

F. Pervasive Computing: Pervasive Computing is considered to be the next generation computing technology. Automatic Spam Filtering of messages is one pf the many applications of filtering in pervasive devices.[13]

G. Location Tracking: The accuracy of Global Positioning System (GPS) is between 10-15 meters. The track obtained by GPS is a sequence of co-ordinates denoting waypoints acquired after specified time intervals or distances. This often results in random errors. To minimize these errors, position filtering is used.[14]

H. UAV: Unmanned Aerial Vehicles (UAVs) are able to connect point cloud data of high resolution. Different types of filtering methodologies are used to filter 3D raw point cloud data collected by UAVs.[15]

I. Cyber Security: Web filtering is a vital aspect of any company's cyber security strategy. It ensures that the users in the network will not be able to access objects infected with viruses. Malicious web content often delivers malware or can be implemented to steal sensitive data and credentials of the users. Web filtering helps reduce these threats.[16]

J. Fog Computing: Nowadays, there is a drift from traditional social networks (IC-SN). Existing security schemes for traditional networks are not suitable for IC-SN. Hence, FCSS, which is a content-aware filtering technique implementing fog computing, is used for security services.[17]

K. 5G and 6G Networks: 5G and 6G networks will have highly advanced performance and network connectivity. The URL filtering techniques currently used can not do real-time filtering. A real-time binary classification model using machine learning will be able to handle URL traffic classifying it into vulgar and clean content instantaneously.[18]

V. Challenges Of Filtering Techniques

The enhancement in SNR caused by filtering is promising; however, filtering often influences the target signal in an undesirable manner. For instance, portions of the desired signal within the stopband range are lost. The situation is made worse by the unwanted distortions experienced by the target signals. Such contortions are contingent on the filter's frequency, amplitude as well as phase characteristics. A filter produces its output by convoluting the input and the impulse response. Each output sample is thus obtained as the weighted sum of multiple input samples. Thus, every output sample is determined by a whole segment of the input, over an extended period of time. In other words, temporal features present in the input signal get spread all over the output signal; again, erroneous features not present in the original signal often end up in the filtered signal.[19] Another underrated disadvantage of filtering is a phenomenon called temporal blurring which means time relationships between a sequence of events often can't be deduced directly from the filtered data. This problem becomes inconsequential if the impulse response of a filter is relatively short in comparison to the observed phenomena. However, that seldom happens.[20]

VI. Filtering Techniques For EEG Signals

The commonly used filters in EEG signal processing can be classified into: high pass, low pass and notch filters. The former attenuates low frequency elements of the input and allows high frequencies to pass across. Similarly, the second one diminishes high frequency activity. The notch filter, on the other hand, removes activity in a very specific range. The alternating current in standard electric outlets usually oscillates at 60 Hz. The electric fields thereby produced frequently contaminate the EEG. To deal with this particular issue, 60 Hz notch filters are implemented.[21] The ideal filter would get rid of all of the electrical noise and artifacts, only allowing true cerebral activity to pass. However, realistically, it is infeasible as filters remove frequencies following strict mathematical rules.

A. Advantages: There are valid reasons behind filtering out certain components of EEG signals on the basis of simple mathematical assumptions. It is safe to assume that the brain generates EEG waves within a certain range of frequencies and that activities beyond that range are possibly not of cerebral origin. It is assumed that activities far below 1 Hz and far above 35 Hz are not cerebral activity and most likely are in fact electrical noise or artifact. However, this assumption is not completely true. Quite often, low frequency noise originates from movement of the head, electrode wires, and perspiration of the scalp while high frequency noise arises from electromagnetic interference and muscle contractions.[22] Based on this assumption that the frequencies of most cerebral activities lie within this bandwidth, EEG filters have one component filter for rejecting most of high-frequency activity and another for low-frequency activity. The range of frequencies passing through is called bandpass of the filter.[23]

Instead of rejecting spurious activity, filtering techniques can also be used to highlight certain EEG activity that would otherwise have been masked by other, high voltage activity. In this case, the electroencephalographer might intentionally attenuate the slow activity in an EEG record despite it representing true cerebral activity, in order to draw attention to fast activity that would normally be lost amidst high-voltage slow waves.[24]

Another reason behind using filters in EEG signal pre-processing is aliasing. Aliasing is a phenomenon where high frequency noise gets representation in the data as low frequency artifacts. For example, during recording, a noise signal at 1700 Hz was present. The EEG amplifier would not detect this since it samples, say, only 500 times per second. Thus,3.4 oscillations would be occurring between two samples recorded by the EEG amplifier. Now, 1700 is not an even multiple of 500; so, each sample will be recording a different phase of this oscillation- sometimes peak, sometimes trough and at other times, somewhere in between. Thus, the EEG signal measured by joining the dots from the different samples, appears to be a much lower frequency oscillation. Because of this phenomenon, there exists a highest frequency that is possible to be accurately recorded at a particular sampling rate. This is termed as the Nyquist frequency. Usually, EEG hardware automatically low-pass filters data at the time of recording EEG, using a threshold set by hardware engineers to fend off aliasing. [25]

B. Issues: Temporally isolated brain events like a neuronal "spike" are represented by one or more than one impulse. Such events become spread out over time after being filtered. Because of this, the event no longer has a clearly defined temporal location. Moreover, a real brain event is never an infinitely narrow impulse response. Its width is finite and hence the filter's response to an event like that differs considerably from the ideal response. If the impulse response features are wide as compared to the studied event, they are identifiable but those which are narrower, might get smoothed away.[26] A sudden oscillatory activity in the brain may be represented in the form of a sinusoidal pulse which after getting filtered, is smoothed out and spread out resulting in temporal blurring. For instance, the pulse thus obtained after filtering might get delayed especially it is a causal filter or might begin even before the event starts if an acausal filter has been used. Such effects become increasingly obvious for filters with a narrow passband.[27]

There are some other common side-effects of filtering. Information belonging to the attenuated frequency range is often lost. Also, high pass filters might hide those fluctuations in the brain potential that are slow. On the other hand, low pass filters might disguise neural activities of high frequency. Some common instances are peaks getting smoothed, steps turning to pulses etcetera. Sometimes, artifacts pop up, for example, response peaks and oscillations called "ringing" produced as a consequence of filtering in reaction to some or the other feature of the original signal. Perhaps, the most nuanced unwanted consequence is temporal blurring which makes the association of brain measurements with stimulation and/or behavior less evident since the temporal relationships between events become far less pronounced in data after filtering. This problem is not significant in case the impulse response of the filter is relatively short compared to the phenomenon under observation . However, in practice, impulse responses of common filters usually last longer than hundreds of milliseconds while significant phases of neural activity have much shorter duration.[28]

VII. Reviews Of Filtering Techniques

This literature review provides a comparative analysis of the efficiency of different commonly used filters. A very important application of filtering will be the denoising of images in image processing. Images contain different kinds of noises arising from sensor faults, lens bends, software artifacts, blurs etc. Tushar Jaiswal *et al* (2009) proposed a system for cancelling noise in images called LFSGW. This detects the bulk of impulse noise points on the basis of the values acquired by the adjacent points that are hereby utilized for cancelling the noise. Images with noise density less than 80 % can be denoised by iterating the algorithm only once. In case of images with noise having density higher than 80 %, not more than two iterations are necessary to create a reconstructed image of best quality.[29] Priyanka Balwinder Singh *et al* (2013) made use of the median filter to de-noise salt and pepper noises as well as Poisson noise from digital images. It was observed that Median filters protect edges and at the same time, remove random noise.[30] Ramalakshmi *et al* (2013) created an anisotropic filter that eliminates background noise while protecting the points on the edges of the images. Here, parallel filtering is used along with contrast stitching.[31] Malik *et al* (2016) suggested a denoising approach which uses an optimization algorithm involving the use of smoothing linear filter and nonlinear filter.[32] Vasudevan *et al* (2019) suggested a similar approach implementing an adaptive median filter using the concept of adaptive fuzzy switching. Henceforth, the filtered output is estimated using Discrete Wavelet Transform and Cuckoo Search Algorithm.[33] Ashokan *et al* (2020) developed a new approach using a bilateral filter under optimized conditions to denoise images without blurring the edges to overcome the challenges posed by over-blurring while smoothing.[34] Himanshu Singh *et al* (2021) modified the Cuckoo Search Optimization algorithm to propose a new methodology for denoising using a guided filter.[35]

Another important application of filters is denoising EEG signals. A considerable amount of study has been conducted in this field in recent years using four main kinds of filters:

A. Moving Average Filter: For this filter, the basic principle is that it first calculates the mean $SMA_{k,prev}$ over the last k data points of a sample of n points as shown in (6) and then calculates $SMA_{k,next}$ from $SMA_{k,prev}$ as shown in (7):

$$SMA_k = \frac{1}{k}(p_{n-k+1} + p_{n-k+2} + \cdots + p_n) \qquad (6)$$

$$SMA_{k,next} = SMA_{k,prev} + \frac{1}{k}(p_{n+1} - p_{n-k+1}) \quad (7)$$

Here, p_{n-k+1} , p_{n-k+2} ,………, p_n denote the first k data points used for calculating the mean $SMA_{k,}$. While calculating the next mean $SMA_{k,next}$, the previous mean SMA_k becomes $SMA_{k,prev}$ and the new point p_{n+1} is used while p_{n-k+1} is dropped.

It uses the Modified Varri Method to segment the EEG signal. It filters and smooths real-time non- stationary EEG signal and gives far better accuracy than other methods (Azami *et al* 2012).[36] It can also remove high and low frequency noise (Popovic *et al* 2016).[37] An optimized moving average filter was proposed by Ferreira *et al* (2016) to remove gradient artifact in fMRI signal.[38] Poor contact of electrodes with the scalp as well as change in impedance in the electrodes, will cause baseline wander in the acquired EEG signal. To remove this, Cascaded Moving Average Filter as proposed by Sreekrishna *et al* (2016) is used which combines two moving average filters having overlapping samples.[39] The performance is rated by parameters like SNR and mean. The COWA filter gives higher SNR in comparison to preexisting methods (Pander *et al* 2019).[40]

B. Adaptive Recursive Bandpass Filter: Gharieb *et al* (2001) separated raw EEG data from sleep spindles data implementing this filter.[41] Jeyabalan *et al* (2007) strengthened the acquired EEG signal using such a filter with autoregressive modeling.[42]Traditional filters like Chebyshev , Butterworth, FIR bandpass filter as well as adaptive recursive bandpass filters are also implemented on EEG signals for removal of artifacts (Machavarapu *et al* 2014).[43] Non-Integer Bandpass filter implements LIRA method (Laguerre Impulse Response Approximation) to remove unwanted noise (Baranowski *et al* 2016).[44]

C. Finite Impulse Response (FIR) Filter: FIR filters are those filters for which the impulse response diminishes to zero in a finite interval of time. In case of a causal FIR filter with discrete time, every term in the output is given by the following equation (8):

$$y[\mathrm{n}] = b_0 x[\mathrm{n}] + b_1 x[\mathrm{n}-1] + \cdots + b_N x[\mathrm{n}-\mathrm{N}] \qquad (8)$$

where n stands for filter order and x[n], y[n], N and b_i denote the input, output, order of the filter and impulse response value at the i^{th} instant respectively.

A low pass FIR filter enhances the SNR and diminishes the noise that often gets superimposed on an EEG signal as observed by Bartoli *et al* in 1983.[45] FIR filter eliminates the baseline drift as well as powerline interference from the EEG signal containing noise. It uses 64 tap FIR which has 16 bit coefficient with a high dynamic range (Wijesinghe *et al* 2014).[46] Bandpass FIR also removes the baseline drift. Factors like mean and standard deviation influence the filter performance. To get a sharp cut-off, we need to increase the order of the filter.

D. Butterworth filter, Chebyshev filters of types I and II and IIR digital filters: These filters have always been commonly implemented to remove undesirable noise. In 2014, Machavarapu *et al* studied IIR filters and concluded that although side lobes are far less in case of these filters in comparison to other filters, the filtered signal is still more susceptible to noises.[47] A Non-recursive IIR filter works as a neuron in the neural network for removing artifacts. 92 % of the blink artifact component is removed from the EEG signal using Non-recursive IIR filter. Performance of a digital IIR filter in a specific application depends on passband ripple, stopband and attenuation. While deciding on which filter would be the best for a particular task at hand, we choose one with maximum slope between the passband and stopband frequencies (Sarkar *et al* 2018).[48] Mahesh Y. Ladekar *et al* (2021) conducted a visual cognitive analysis of workload based on EEG signals with the help of multirate IIR filters.[49] .

In 2000, G. Alaron *et al* used a Butterworth filter having three poles in electroencephalography using an algorithm simple enough to compute filter coefficients taking any random frequency as cut-off with ease.[50] In 2009, Li Ke et al used Chebyshev type I filter as well as Multiscale filtering to classify EEG signals.[51] In 2014, Hasan *et al* used a low pass 4th order Butterworth filter in designing an inexpensive wireless EEG Acquisition System.[52] E. Juárez-Guerra *et al* achieved an accuracy of 99.26 % using Digital Chebyshev type II and Digital Elliptic in Epilepsy Seizure Detection from EEG signals, in 2014.[53] Jinn Jai *et al* , in 2014, used Chebyshev filters of types both I and II along with Butterworth filters for using EEG to automatically detect drowsy state. It was observed that for any particular filter specification, for instance, pass-band or stop-band ripple or sampling frequency, lower order filters are sufficient for both types of Chebyshev filters in comparison to Butterworth filters.[54] Md Belal Bin Heyat *et al* in 2015 used a Butterforth filter of order 4 having inferior cut-off frequency equal to 0.5 Hz and superior given by 35 Hz to riddle EEG signal for wireless transfer.[55] S.S. Daud *et al* in 2015 compared Butterworth Bandpass filter of order 4 to Stationary Wavelet Transform. The latter was found to be more efficient in eliminating noise while retaining the original signals[56] Shakshi *et al* used 2nd order Butterworth Bandpass filter to overcome the problems created by noise, artifacts and powerline components during classification of brainwaves as well as extraction of relevant features from EEG signals, in 2016.[57] In 2016, Manjit Sandhu et al studied Butterworth, Chebyshev and Elliptic filter and it was concluded that Butterworth filter happens to be the best trade-off considering attenuation as well as phase response and moreover, this is devoid of ripples.[58] In 2017, Evi Septiana Pane *et al* used Bandpass IIR filter having a window of Chebyshev type II and achieved an accuracy of 81.64% while separating EEG signal into different bands: alpha, beta, gamma and theta.[59] In 2018, R. Widaldi *et al* used Elliptic Highpass filter to allow only the gamma band of EEG signal to pass through.[60] Rayhan Habib Jibon *et al* used two IIR filters namely a Notch Filter as well as a Chebyshev filter of type II to eliminate the inference introduced by power transmission lines.[61] In 2019, Navid Ghassemi *et al* used high pass Butterworth filter to remove low frequency noises from signals, while detecting Epileptic seizures. [62] Tuerrun Waili et al in 2019 used Butterworth filters- highpass as well as lowpass- to determine whether the emotional state of a person is stressed or calm from the filtered EEG signal. [63] In 2020, Prajoy Podder *et al* studied the impulse , magnitude and phase response of IIR filters like Butterworth, Chebyshev type I and elliptic filter of the following types: low pass, high pass, band pass and band stop, while filtering speech signals and deduced that Butterworth filter was the best trade-off keeping both phase response and attenuation in mind.[64] P. Nagabushanam *et al* , in 2020, used Elliptic low pass filter of passband between 0-40 Hz with amplitude of ripples lower than 3 dB in the passband for artifact and power line noise removal while constructing a classifier for EEG signals. [65] Montree Kumngren *et al* , in 2020, designed a Butterworth lowpass filter of 0.5 V and of order 5 using OTA with Multiple Inputs to be used for ECG, reducing the number of components from 8-10 to 5.[66] In 2021, Fahd A. Alturki used bandpass Butterworth filter of order 5 for removal of interference as well as noise produced by electrodes in addition to magnetic fields generated by nearby devices while using EEG to diagnose conditions likeAutism, Epilepsy, etcetera.[67] M Thilagaraj *et al* used Butterworth filter for identifying drivers' drowsiness on the basis of features extracted from EEG signals.[68] In 2022, Yilong Zhu *et al* designed a Quasi-Elliptic filter that was compact and had a Dual-band, using the principles of Hybrid SIW along with Microstrip Technologies.[69]

Besides these four types of filters, some others have also been applied in the sector of Biomedical Signal Processing. Kritiprasanna Das *et al* (2021) used multivariate iterative filtering for Schizophrenia detection from multichannel EEG signals. [70] Harishvijey et al (2022) devised an automated method to process EEG signals in order to detect seizures using a Gaussian filter with optimized variables.

TABLE 1:SURVEY OF LITERATURE

Sl. No.	*Year*	*Ref.*	*Technique Used*	*Inference*
1	2000	[50]	Digital Butterworth filter with three poles and any random cut-off frequency	Simplicity of algorithm while calculating filter coefficients makes it useful for electrophysiological applications
2	2009	[51]	Chebyshev Type I, Multiscale filtering	Chebyshev Type I produced a maximum accuracy rate of 87.71% whereas multiscale filtering achieves an accuracy rate of 91.13%
3	2012	[36]	Moving Average Filter and Savitzky-Golay filter	Improved performance than simple Modified Varri method though not as good as Savitsky-Golay filter
4	2014	[46]	0.5 Hz highpass and 50 Hz notch Finite Impulse Response (FIR) filter	Baseline drift as well as interference by powerline components removed; dynamic range maximized
5	2014	[43]	IIR digital filters-Chebyshev filter, Butterworth, FIR bandpass and Elliptic filters	Elliptic Filter achieves highest accuracy of 84.3% in case of training data and an accuracy of 86.45% in case of testing data- but fails in achieving 100% accuracy for unsupervised conditions
6	2014	[52]	Butterworth filter	A low pass Butterworth filter of order 4 is applied to eliminate unwanted high frequency signals.

7	2014	[53]	Chebyshev II along with Elliptic Filter	Average accuracy of 99.26 % was obtained
8	2014	[54]	Chebyshev type I and II along with Butterworth filter	For a particular filter specification like passband ripple, stopband ripple and sampling frequency, Chebyshev filter of either type of lower order is sufficient compared to Butterworth filter which will require higher order
9	2015	[55]	Butterworth filter	The EEG signal is riddled by Butterworth filter of order 4 with lower and higher cut off frequencies given by 0.5 Hz and 35 Hz respectively
10	2015	[56]	Butterworth Bandpass Filter	EEG signal is filtered by Butterworth Bandpass Filter of order 4 as well as Stationary Wavelet Transform and the latter is found to be more efficient in removal of noise
11	2016	[44]	Non-integer bandpass filters using Laguerre Impulse Response (LIRA) method	Less distortion of signal; however, the method is not always convergent
12	2016	[39]	Real time cascaded moving average filter	Effective in removing baseline wander; used for detrending EEG signals
13	2016	[37]	Median filter applied on sEMG , median filter applied on fractional derivative of sEMG and Moving average(MA) filter	Artifact performance for Median filter is better than MA filter; fractional derivatives reduce amplitudes of ECG peaks in sEMG envelopes obtained with median filter; procedure isn't applicable to real time processing because of delays
14	2016	[57]	2^{nd} order Butterworth Bandpass filter having cut-off frequency 0.5-60 Hz is used.	Use of FIR Butterworth filter helps in overcoming the problems created by noise and artifacts and powerline components
15	2016	[58]	Butterworth, Chebyshev and Elliptic filters	Butterworth filter forms the best trade-off between phase response and attenuation and moreover, is devoid of ripples.
16	2017	[59]	IIR Bandpass filter with a window of Chebyshev type II	Accuracy of 81.64% was obtained while separating EEG into the following bands: gamma, beta, alpha and theta
17	2018	[60]	Elliptic Highpass filter	Elliptic Highpass filter passes the gamma band EEG signal.
18	2018	[61]	IIR Notch Chebyshev type II filters	IIR Notch and Chebyshev type II filters used to eliminate interference introduced by power transmission lines.
19	2019	[62]	Butterworth filter	Low frequency noise is removed from signals by use of high pass Butterworth filter.
20	2019	[63]	Lowpass and Highpass Butterworth Filter	The data was isolated by using a highpass and a lowpass Butterworth filter to filter the signal
21	2019	[40]	Cascaded ordered weighted aggregation (COWA) filter	Achieves reduction in interference which is caused by impulses and can not be suppressed efficiently by linear filters; significant improvement in SNR; however only two-layer OWA filter has been studied
22	2020	[64]	Butterworth, Chebyshev type I and elliptic lowpass, highpass, bandpass and bandstop filters	Butterworth filter is found to be the best trade-off between phase response and attenuation
23	2020	[65]	Elliptic lowpass filter of passband range of 0-40 Hz having ripple <3 dB	The classifier obtains an accuracy equal to 80.00% while training and 73.33% during testing
24	2020	[66]	Butterworth Low-Pass Filter of order 5 and voltage equal to 0.5 V	Use of Multiple Input OTA in case of ECG Applications reduces the active elements required to realise the filter from 8-10 to 5
25	2021	[67]	Multirate IIR filter	Computational complexity using IIR filter reduces by 70% and 45% in comparison to Morlet and analytic wave transform respectively
26	2021	[70]	Multivariate iterative filtering	Multivariate filtering essential for multichannel data;the highest accuracy achieved by SVM classifier is 98.9%
27	2021	[67]	Butterworth filter	5^{th} order bandpass Butterworth filter used for removal of noise and interference produced by electrodes as well as magnetic fields created by neighbouring devices
28	2021	[68]	Butterworth filter	Maximal flatness in frequency response obtained in the pass band which does not contain any ripple and it approaches zero in the stop band; also more linear phase response obtained in comparison to Chebyshev and Elliptic filters
29	2022	[69]	Quasi Elliptic filter with Compact Dual Band using Hybrid SIW as well as Microstrip Technologies	High selectivity and the Transmission zeros can be controlled ; compact in size; coupling paths for each Substrate Integrated Waveguide are independent
30	2022	[71]	Gaussian Filter	Value of performance metrics obtained are 98.48%, 98.44% and 98.51%

IX. CRITIQUES

It is observed that in the last couple of years, significant amount of research has been done in Brain Science using common IIR filters in order to filter EEG. Here, we have considered 30 papers from 2000 to 2022 where IIR filters have been used. The following graph displays how frequently

Butterworth, Chebyshev type I, Chebyshev type II and Elliptic filters have been implemented in the papers studied above. It can be inferred that Butterworth is the most commonly used filter followed by Elliptic, Chebyshev type I and Chebyshev type II respectively.

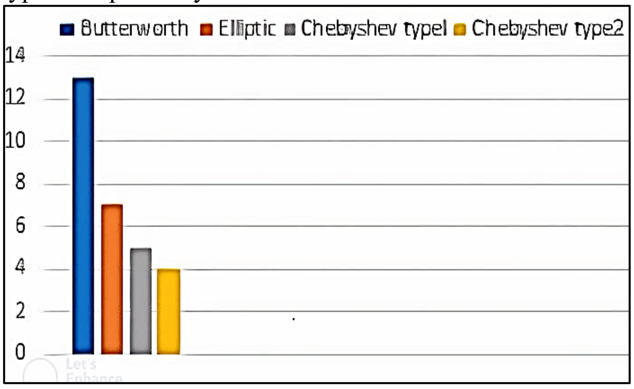

Fig 1: Frequency of use of common IIR filters in the papers studied

X. Conclusion

It is often necessary to have circuits which are capable of selectively filtering a particular frequency or a range of frequency out of a mix of different frequencies. An electronic filter is a circuit designed for this specific purpose. In this paper, a general overview of different kinds of electronic filters and their applications have been given along with special emphasis on filtering techniques in EEG signal processing. Out of all common IIR filters, Butterworth filter has been found to be most predominantly used. Butterworth filter does not have peaking which sometimes makes it a better choice in control systems.[72] In our future works, special emphasis will thus be given on Butterworth filters and their application in signal processing in Brain Science. Furthermore, the application of new techniques like the Fuzzy filter, the Probor filter, etc. can enable us to improve the accuracy of the results attained.[73] Other techniques like Bypassing and Post-Regulation can also be implemented simultaneously with our choice of filter to achieve greater overall performance efficiency.[74]

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